WORKSHEET

Chapter 3: Mixed sequences and series

1 Classify each of the following sequences as arithmetic, geometric or neither.

a 3, 3, 6, 9, 15, 24, …

b 5, −15, 45, −135, 405, …

c 1, 3, 6, 10, 15, 21, …

d $7, 5\frac{2}{3}, 4\frac{1}{3}, 3, 1\frac{2}{3}, \ldots$

2 Find a rule for the nth term of each of the following sequences.

a 3, 5, 7, 9, …

b 20, 19, 18, 17, …

c 1, 2, 5, 26, 677, …

d $\frac{3}{7}, \frac{6}{7}, 1\frac{5}{7}, 3\frac{3}{7}, \ldots$

3 Find the sum of the first seven terms of each of the following sequences.

a $\frac{2}{9}, \frac{2}{3}, 2, 6, \ldots$

b 11, 7, 3, −1, …

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4 Find the sum of each of the following series.

a $\frac{3}{4} + 1\frac{1}{2} + 3 + \ldots + 12\,288$

b $23 + 18 + 13 + \ldots - 72$

5 Find the first four terms of:

a an arithmetic sequence with $t_5 = 33$ and $t_{11} = 57$

b a geometric sequence with $t_6 = 1610.51$ and $t_9 = 2143.589$

c an arithmetic sequence with $t_{13} = 28$ and $t_{20} = 77$.

6 Find the sum of the first 10 terms of:

a a geometric sequence with $t_3 = 1000$ and $t_7 = 3.906\,25$

b an arithmetic sequence with $t_5 = 35.8$ and $t_{21} = 47$

c a geometric sequence with $t_2 = 3$ and $t_9 = 6561$.

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Answers

1 **a** Neither

b Geometric

c Neither

d Arithmetic

2 **a** $2n - 1$

b $t_n = 21 - n$

c $t_n = (t_{n-1})^2 + 1, t_1 = 1$

d $t_n = \frac{3}{7} \times 2^{n-1}$

3 **a** $242\frac{8}{9}$

b -7

4 **a** $24\,575\frac{1}{4}$

b -490

5 **a** 17, 21, 25, 29

b 1000, 1100, 1210, 1331

c $-56, -49, -42, -35$

6 **a** 21 333.313

b 361.5

c 29 524

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